

Circuit Level Phase Dimming with Avi-on Networks



ORDERING INFORMATION

To order please contact Avi-on sales at **(877) AVION-US**, (877) 284-6687 or prosales@avi-on.com for information on becoming an Avi-on partner and order details.

Project	Location/ Type
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SPECIFICATIONS

Input Voltage:	120 VAC	Radio Frequency:	2.4GHz
0-10V Dimming:	Load Current Sink	Wireless Standard:	BLE 4.2 with Mesh
Load Capacity:	1 W min, 600W Max Incandescent/ Halogen Recommended 300W Max LED and Other	Point to Point Range*:	80ft with obstructions and 350ft unobstructed
Size:	4.00"H x 4.00"W x 1.81"D with 0.50" NPT nipple (102mm x 102mm x 46mm)	Warranty:	5 years; 10 years optional
Mounting:	External or internal box mount	Regulatory:	FCC: 2AFZI-AV1010 B IC: 20544-AV1010 B BQB: D031801, DID: 86303 UL 916, C-UL, NEMA 1
Weight:	8oz (16g)		
Operating Temperature:	-30F to +122F (-34C to +50C)		
Storage Temperature:	-40F to +185F (-40C to +85C)		
Humidity Rating:	95% non-condensing		

**When communicating through the mesh, range is essentially unlimited (5000ft+)*

INSTALLATION AND CONFIGURATION

- 1) Verify that the connected load does not exceed the device ratings.
- 2) Verify the dimming type of the connected fixtures and set the Forward/Reverse Phase DIP switches to match the fixture type being controlled. Do not mix different fixture types on the same dimmer. **IMPORTANT:** Leave the Current Source 0-10V DIP switch set to OFF.
- 3) Install the XFAC and Phase Dimmer in a 4x4 or equivalent electrical box following local electrical code requirements. Connect the AC, DC power, and 0-10V Dimming wires as shown in the wiring diagram below. Cap off the Purple/White wire.
- 4) When configuring the XFAC in the Avi-on network, **Low End Trim must be set to 25%** to avoid potential fixture flicker and to ensure expected operation of wall stations, daylight harvesting, and sensors.
- 5) The unit contains a thermal protection cutoff. If the unit is overloaded, it will shut off and restart when it cools down. It is not rated for continuous overloading, however. Overloading can easily lead to premature failures and is not covered by warranty.

Note: Switching on the unit will change the dimming mode between leading edge and trailing edge. A power cycle is required.

The purple and pink wires are used with both current sink 0–10V controllers and current source 0–10V controllers. The AVI-KIT-PHASE-DIM will automatically detect the controller type.

DIMENSIONS & WIRING DIAGRAMS

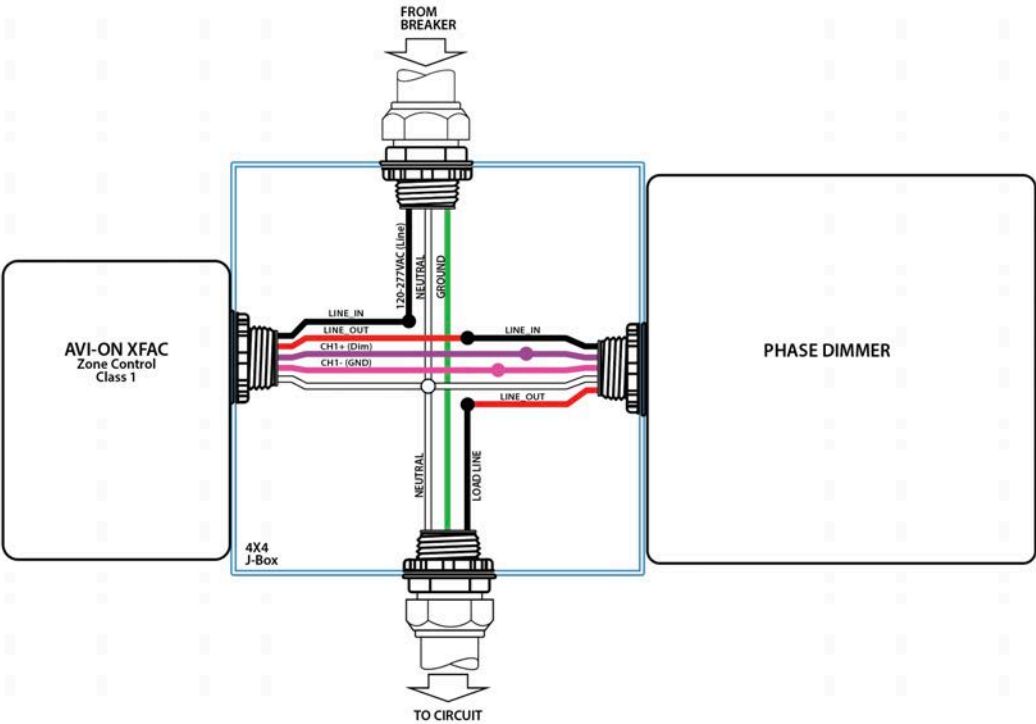


Figure 1. Wiring Diagram

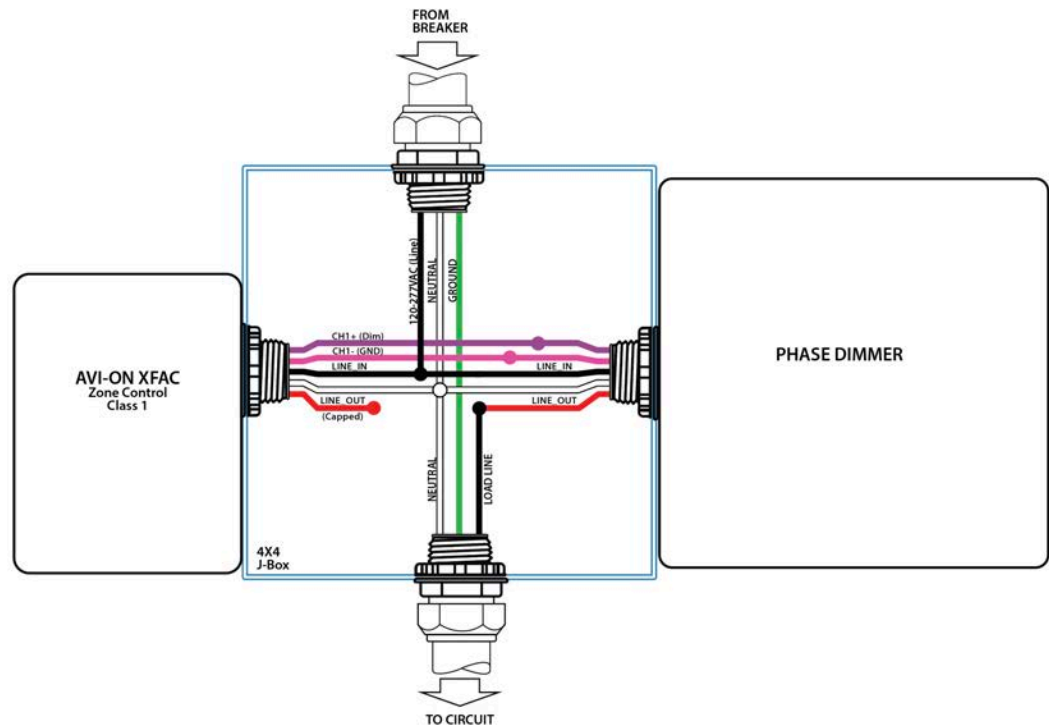


Figure 2. Wiring Diagram

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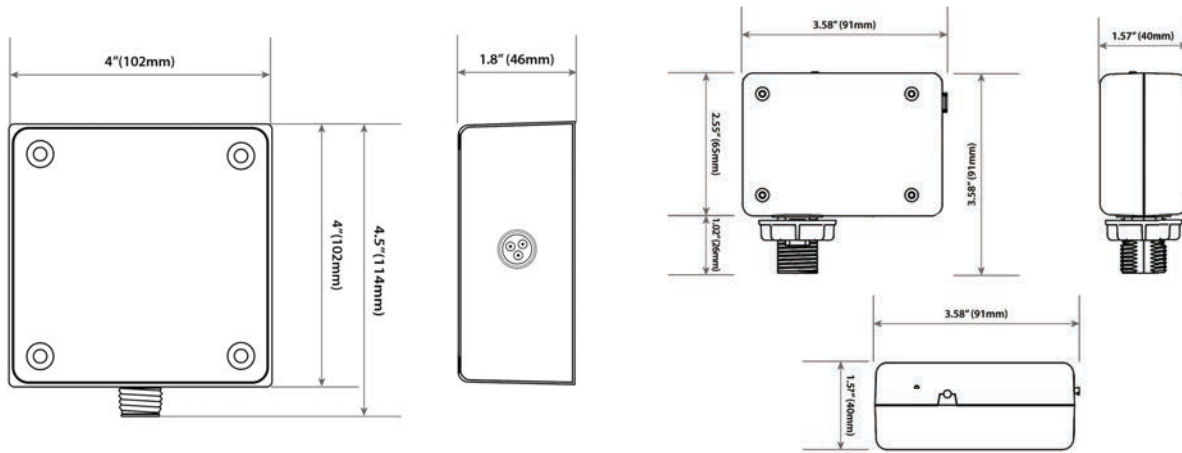


Figure 3. Dimensions

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